

chap1

- how do we define natural number
- how to convert number between base 10 and n. make sure know how to use calculator to do it.
- show that any $\frac{p}{q}$ can be written as finite decimal expansion if and only if Q is divisible by 2 or 5
- Prove that all rational numbers can be written as finite decimals or repeating decimals.
- what algebraic number, what is transcendental number
- If a,b are algebraic number, so are $a + b, a - b, ab$
- let $P(Z) = a_n x^N + \dots a_1 x + a_0$ with coefficient in $\mathbb{C}$, then $P(Z) = 0$ must have n root.
- refer to the concept about quaternion
- **exercise revice**
 - chap1
 - 1(1.1),4(1.3),5,6(1.3)

Chap2

- $\deg(f) = m, \deg(g) = n$
 - $\deg(f + g) = \max\{m, n\}$
 - $\deg(fg) = m + n$
- prove the division, given $f(x), g(x), g(x) \neq 0$, there are unique $q(x), r(x)$ such that

$$f(x) = q(x)g(x) + r(x)$$

where $\deg(r(x)) < \deg(g(x))$ or $r(x) = 0$, can just prove the uniqueness.

- Find the remainder when $x^{99} + x^2 - 1$ is divided by $x^2 - x + 1$
- Prove $p(x) = a_n x^n + \dots + a_1 x + a_0$ be a degree n polynomial with integer coecients (namely, $a_n, a_{n-1}, \dots, a_1$ and a_0 are integers). If $ax - b$ is a factor of p(x), where a and b are relatively prime integers, show that a is a factor of a_n and b is a factor of a_0 .
- prove that if

$$f(x) = q(x)g(x) + r(x)$$

the gcd of f,g also equal to g,r

- prove the bezout identity for polynomial.
- If $f(x)$, $g(x)$ and $h(x)$ are polynomials such that $f(x)$ divides $g(x)h(x)$, and $f(x)$ and $g(x)$ are relatively prime, prove that $f(x)$ divides $h(x)$.
- Prove that if $f(x)$ and $g(x)$ are relatively prime polynomials, and $h(x)$ is a polynomial with $\deg(h) < \deg(f) + \deg(g)$, there exist two unique polynomials $a(x)$ and $b(x)$ with $\deg(a) < \deg(g)$ and $\deg(b) < \deg(f)$, such that

$$h(x) = a(x)f(x) + b(x)g(x)$$

- write the definition of Lagrange interpolation
- Find polynomial $f(x)$ of degree at most 5 such that $f(2) = 69, f(1) = 7, f(0) = 1, f(1) = 3, f(2) = 11, f(3) = 101$.
- **exercise revice**
 - 2.2
 - 2,4,9,10,11
 - 2.3
 - 3,4,6,7,8
 - 2.4
 - 6,7
 - 2.5
 - 4

Chap3, 4

- $\lim_{n \rightarrow \infty} n^k x^n = 0$
 $\forall k, |x| < 1$
- write the definition of binomial expansion
- explain how partial fraction work.
- **exercise revice**
 - 3.1
 - 2,7c,11,12
 - 3.2
 - 4,5
 - 3.3
 - 6,8,9
- **exercise revice**
 - 4.1
 - 1,3

Chap5

- what is complex conjugate of $a + bi$
- what is modulus of a complex number
- $|z|^2 = z\bar{z}$
- what is argument of a complex number, what is principle argument
- what is the trigonometric form of a complex number
- write the Euler's formula
- what is the exponential form of a complex number
- write down the procedure of find the formula of $\cos(nx)$, $\sin(nx)$
- write the procedure of finding the image of P after a counter-clockwise rotation by θ with respect to A.
- write the Demoivre's theorem
- what means w is the n —th root of z
- find the fifth root of $-32i$ and represent them on an argand diagram
- Prove that if $f(x)$ is the real coefficient polynomial, and z is a root, then $\bar{z}$ is also a root.
- what is fundamental theorem of algebra
- write the vieta's formula
- how to use calculator to calculate Re and Im, and argument.◦◦◦
- **exercise review**
 - 5.3
 - 6, 7, 9, 13, 14, 15, 16
 - 5.4
 - 3, 5, 9, 12, 14, 15

Chap6

- what is translation, Find the new coordinates of P(6, 5) after a translation of axes to a new origin at (2, 4).
- write the formula of rotation under a anti-clockwise rotation angle θ with respect to the origin
- A plain curve can be describe by a pair of parametric function,

$$x = f(t), y = g(t), t \in [a, b]$$

what is initial point, terminal point, what means curve is closed, what means curve is simple.

- derive the "万能公式".
- how to define eccentricity.
- what is focus, what is directrix of the conic.
- what is the conic when eccentricity is in different value.
- what is the standard equation of a parabola, where is focus, directrix, vertex of the conic.
- what is the standard equation of ellipse, how to calculate eccentricity, where are foci, where are directrix, where are vertices, where is center, where is major axis and minor axis, what is major diameter, what is minor diameter.
- what is the standard equation of hyperbola, how to calculate eccentricity, where are foci, where are directrix, where are vertices, where is center, where is major axis and minor axis, what is major diameter, what is minor diameter. How to calculate asymptote.
- If P is a point on an ellipse, and F and F' are the foci, then
$$|PF| + |PF'| = 2a$$
- If P is a point on a hyperbola, and F and F' are the foci, then
$$||PF| - |PF'|| = 2a$$
- For general second degree equation

$$Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$$

what θ should choose to rotate for eliminating the xy terms.

- For general second degree equation

$$Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$$

what does $B^2 - 4AC$ represent

- derive the formula of tangent line equation, namely for ellipse, the tangent line equation at (x_0, y_0) is

$$\frac{x_0x}{a^2} + \frac{y_0y}{b^2} = 1$$

- **exercise review**

- Varberg

- 10.4

- 58

- 10.1

- 15, 30, 32, 37, 38, 39

- 10.2
 - 43, 51
- 10.3
 - 29, 35, 39, 41, 42, 46, 67
- Stewart
 - 10.5
 - 33, 45, 48, 57

Chap7

- what is polar coordinate
- what is limacons, what is cardioids
- what is lemniscates
- $\cos \theta \rightarrow \sin \theta$ turn $+90^\circ$
- what is roses, how to determine how many leafs.
- what is a spiral of Archimedes
- what is the polar equation of a line
- what is the polar equation of a circle if it pass through pole.
- what is the polar equation of conic with one of the focus at pole.
- **exercise revice**
 - stewart
 - 10.3
 - 11, 18
 - 10.6
 - 1, 3, 5, 7, 13
 - Varberg

Chap8

- the distance formula
- midpoint of $(x_1, y_1, z_1), (x_2, y_2, z_2)$
- what is magnitude
- what is dot product
- what is direction angles
- what is the scalar projection and the vector projection of a on b
- write the definition of cross product

- $a \times (b \times c) = ?$
- $(a \times b) \cdot c$
- notice that

$$\begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & a_2 & a_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = a \cdot (b \times c)$$

- what is the volume of the parallelepiped determined by the vectors a , b and c
- how to show three vector in $\mathbb{R}^3$ are coplaner, give me two different aspect. another from linear algebra.
- how to define a line, then what is the symmetric equation of this line.
- write the definition of skew line.
- how we define a plane
- how to find Find the symmetric equations of the line of intersection of two planes, namely how to find the direction vector and also the point that line passed.
- how to find the angle between two plane.
- the formula of Distance from a point to a plane
- how to find distance of two skew line.
- what is cylindrical coordinate, write the transformation between cartesian coordinate.
- what is spherical coordinate, write the transformation between cartesian coordinate.
- **exercise revice**
 - stewart
 - 12.1
 - 15
 - 12.5
 - 31, 51
 - verberg